

VEIR HTS Power Systems: A Reliability, Availability & Maintainability Study

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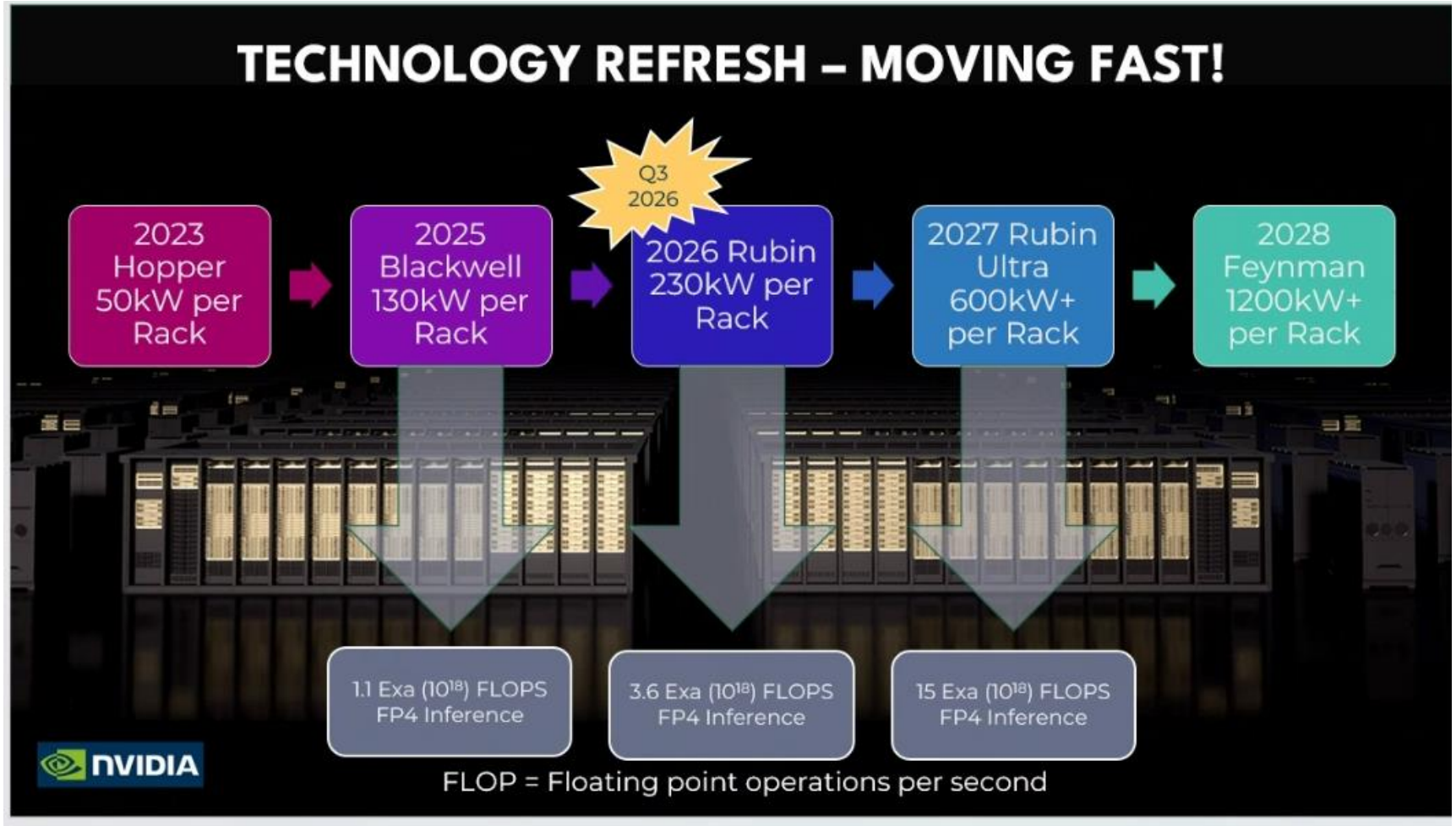
VEIR Inc.



Why This Study

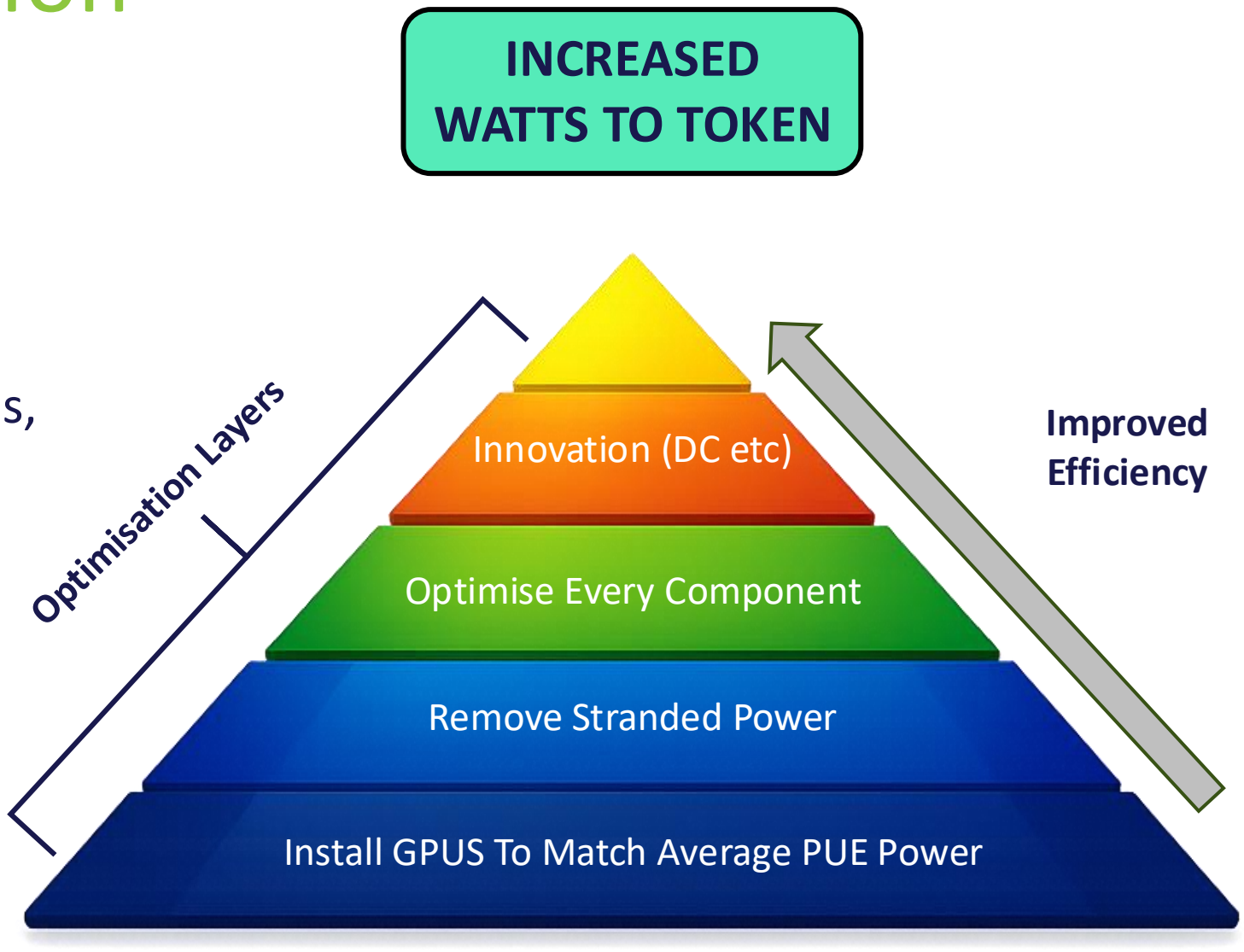
- AI load growth is exceeding the limits of copper distribution.
- VEIR HTS delivers ~10× power density, near-zero losses & a smaller footprint.
- This study quantifies system reliability by iterative topology optimization.

NVIDIA Rack Roadmap



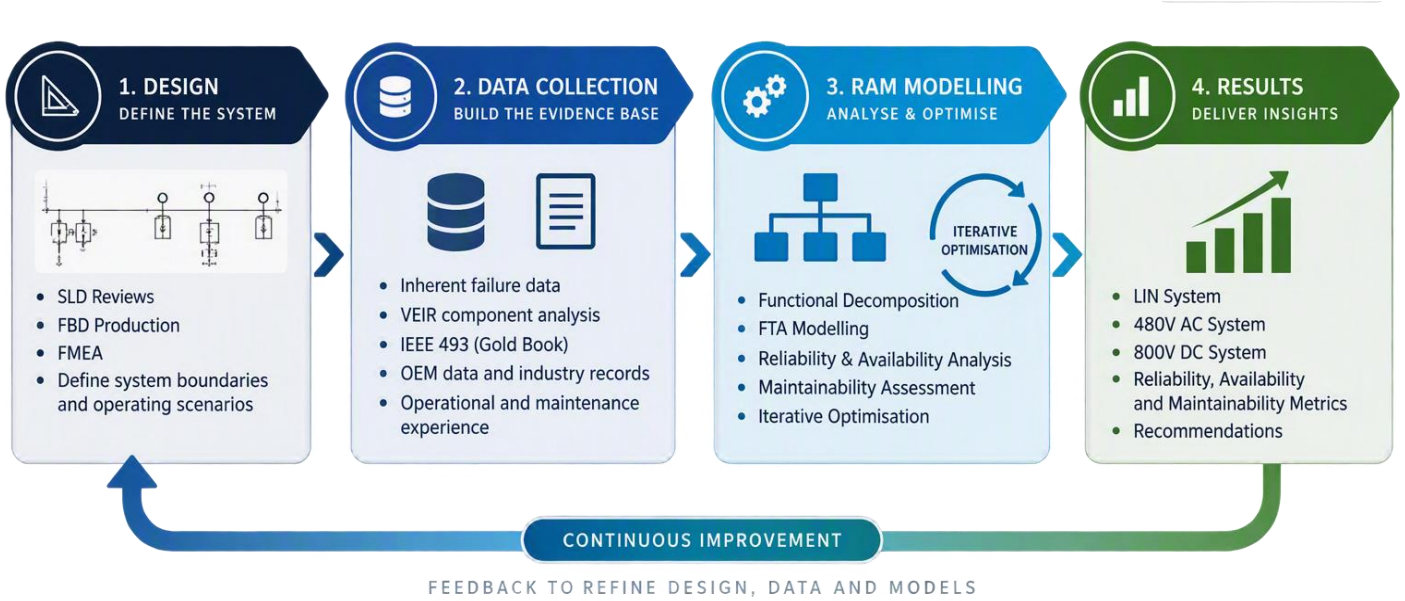
Technology Introduction

- Physics of Superconductors
  - Cryogenic temperature enabled by LN
  - More electrons flowing
  - Strong magnetic field generation
- HTS cables = more power, more tokens, less losses
- Future opportunities
  - Shared Cryogenic infrastructure with Superconducting quantum computers



RAM Methodology

- Model-based RAM approach using fault tree analysis & functional modelling
- Simplified “black-box” representations used for subsystems still evolving
- Iterative Optimization:
  - build baseline model
  - identify single points of failure
  - adjust assumptions/architecture
  - model and compare results
- Data sources: IEEE 493-2007 (“Gold Book”), OEM equipment data, RED Engineering internal reliability statistics

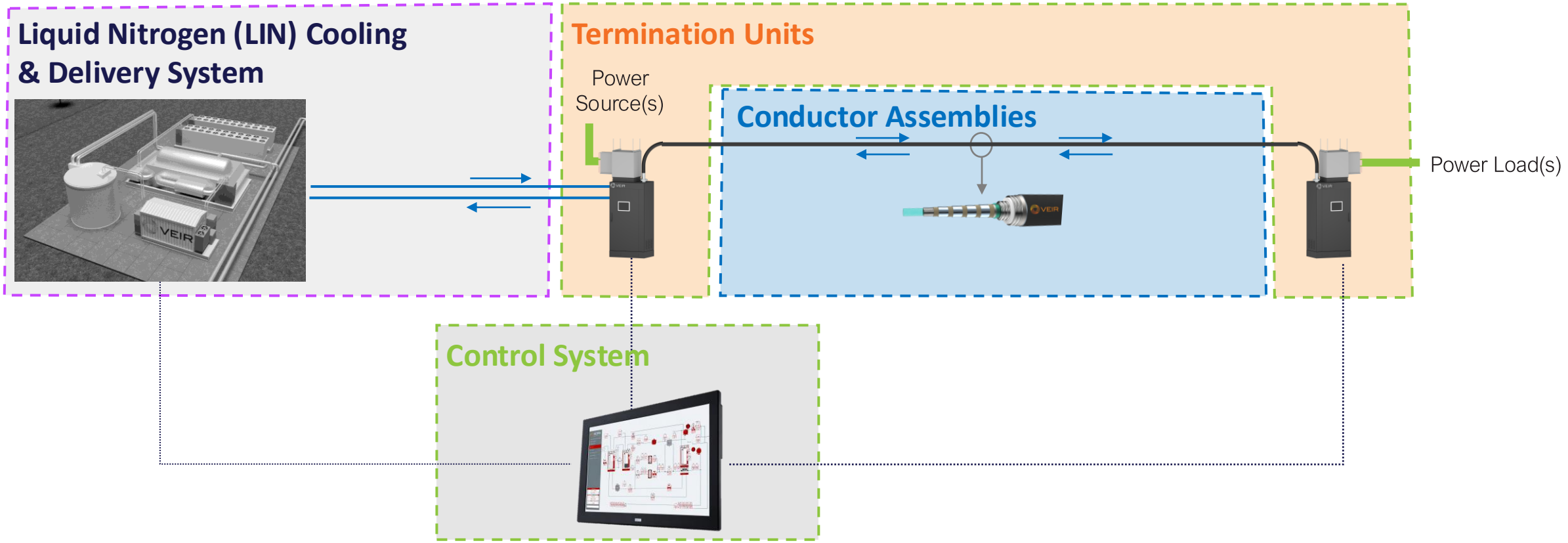


Scope of Work

- Three interdependent systems were assessed:
  - Liquid Nitrogen (LIN) cooling system
  - AC HTS power system
  - DC HTS power system
- For each system, the study examined:
  - Failure modes (MTBF) and dependencies
  - Repair assumptions (MTTR)
  - Built-in redundancy
  - Overall system availability

VEIR SL LV Power Distribution Solution

Purpose-built modularity across four core subsystems enables flexibility and scalability. Standard building blocks drive rapid deployment, quality, and reliability.



Liquid Nitrogen (LIN) Cooling System

SUBSYSTEM FUNCTION

Refrigerates LIN to hold the HTS cable at operating temperature.

Manages closed-loop subcooled LIN circulation, matching energy consumption to system load.

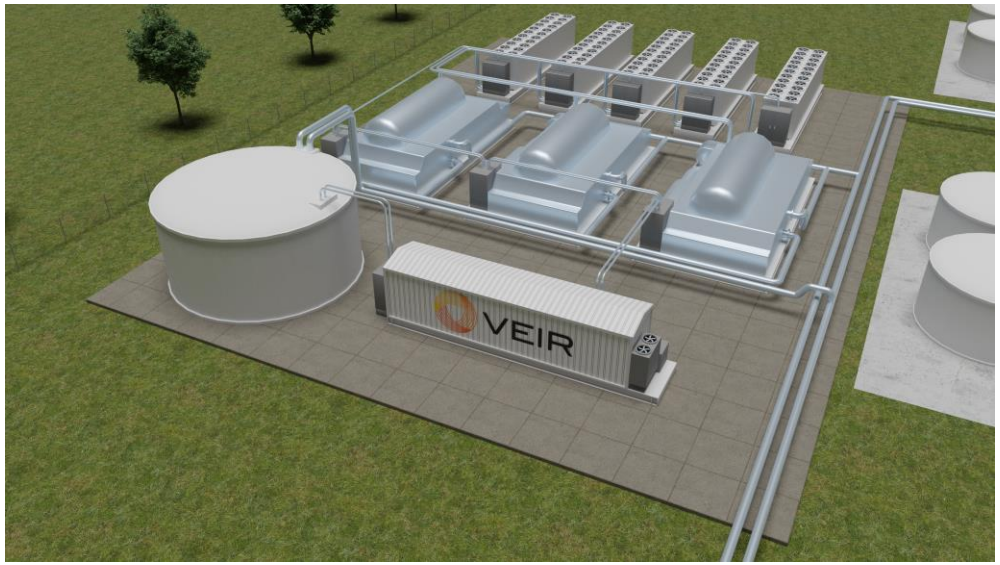
Maintains thermal stability across the full cable length by actively removing heat from pipe insulation and AC losses.

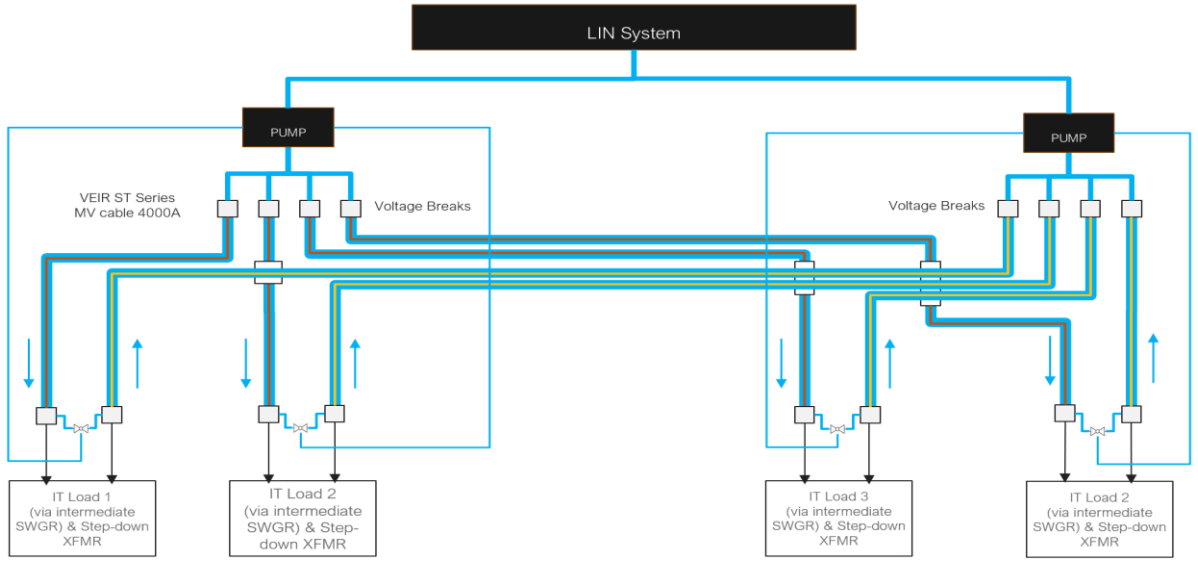
KEY VALUE DRIVERS

Refrigeration power scales linearly with load — no wasted energy at partial load.

Modular cooling plants right-sized to cable length and power rating for each data-center layout.

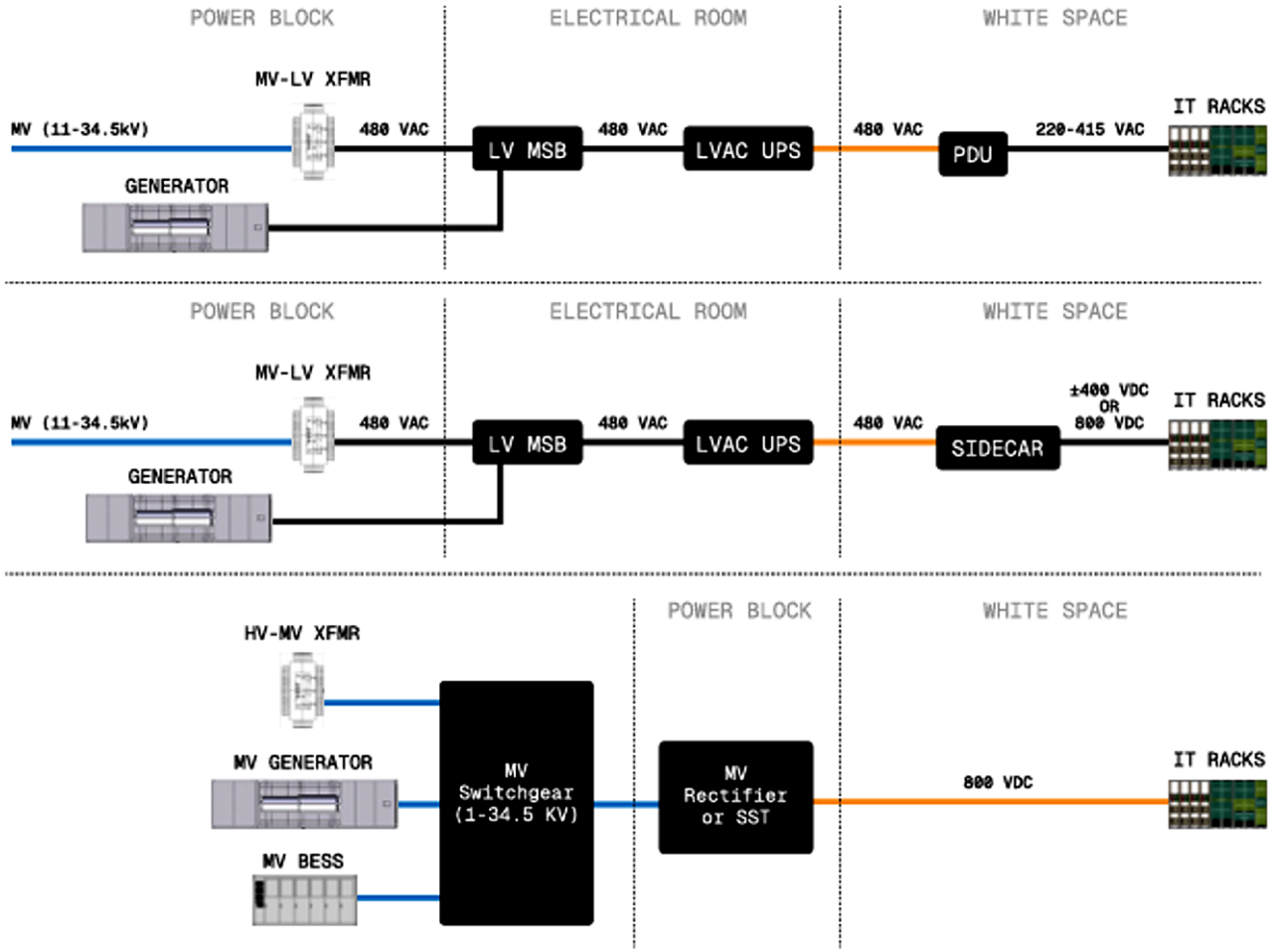
Redundancy optimized configurations can deliver 99.999%+ cooling availability.





Power Distribution Architectures Considered

- HTS LV Power Distribution Architectures considered:
  - AC power block/white space feeder application
  - DC SST to rack LV feeder
- Redundancy Strategies Considered
  - Distributed Redundant
  - Block Redundant



Data Halls : Block Redundant

- **AC: 21 MW IT** capacity|**7+1** block redundant

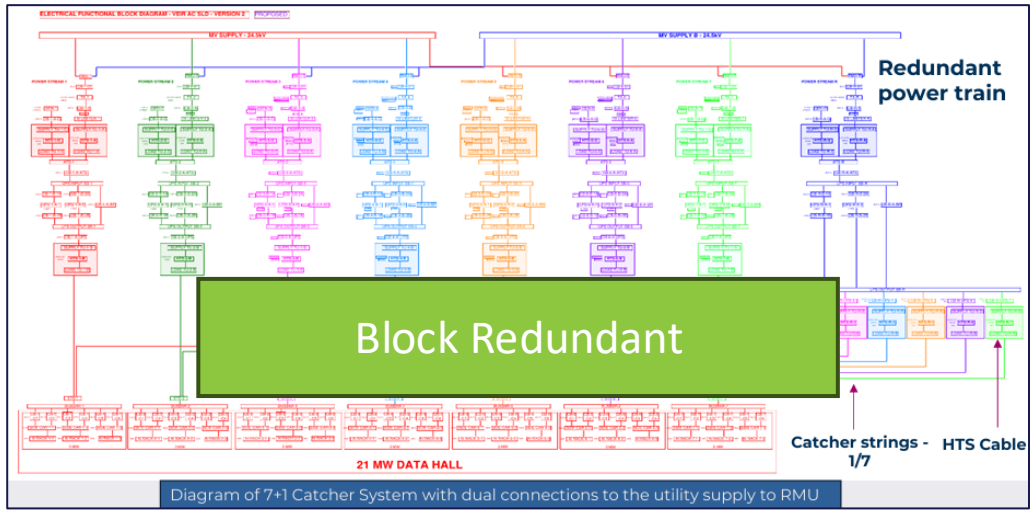
(7 rows each made of 3no. 1MW GPU racks | 1.1 MW sidecar per rack)

- **DC: 20 MW IT** capacity|**2+1** block redundant

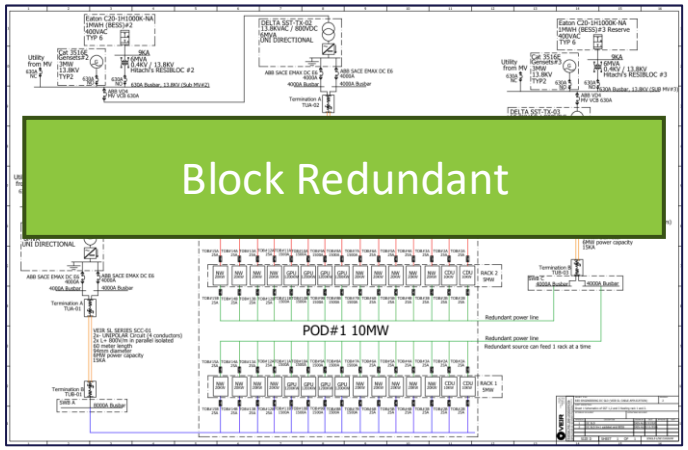
(2no. 10MW pods, each with 2no. 5MW IT rows)

- Each row is fed from a dedicated primary power string
- DC: two primary strings are supported by one redundant string able to back up either row

AC Block Redundant Architecture



DC Block Redundant Architecture



Data Halls : Distributed Redundant

- **AC: 24.3MW IT load**

 - 6 rows (4 MW IT load),

 - 4 power blocks, 28 AC HTS cables

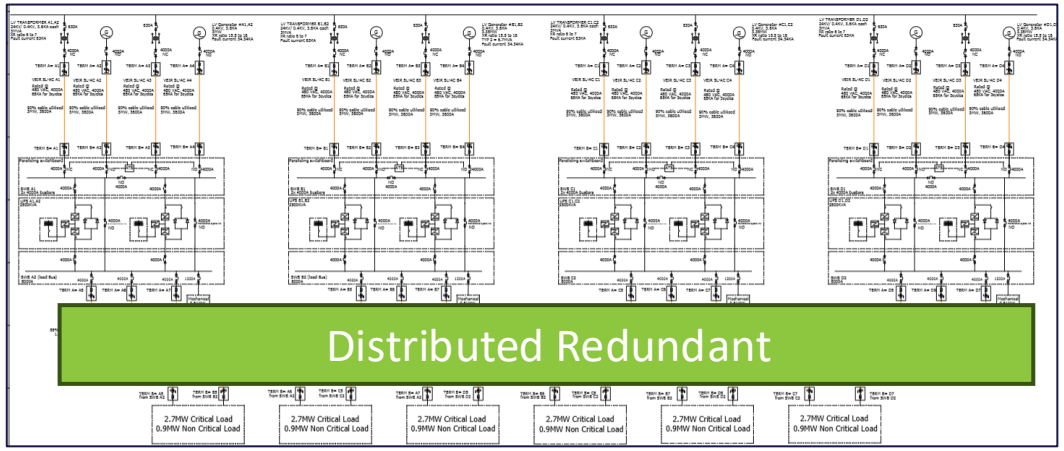
- **DC: 60MW IT load**

 - 6x10MW HACs, 2x5 MW rows /HAC

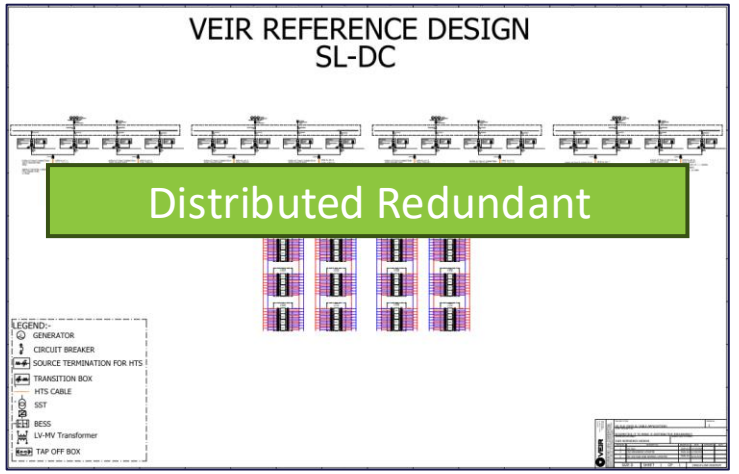
- 8 DC HTS cables in a “4-to-make-3” HTS cable topology

- **DC HTS cables** operate at 75% capacity, ramping to 100% if one cable fails

AC Distributed Redundant Architectures



DC Distributed Redundant Architectures



Results : Block Redundant

- **AC:** generator-backed strings plus one shared redundant string at low voltage - covers only one primary failure at a time
- **DC:** N+1 redundancy at system level, BESS backup at MV, includes rectifiers/SSTs with DC changeover switching – (DC –STS product still in development)
- In both cases, the LIN system is fed from a generator backed UPS power feed for primary power and backed up as follows: Utility & UPS (AC architecture) or utility connection & BESS (DC architecture).
- **Results:**
  - LIN System: 5-nines (99.999%)
  - AC: 3-nines / Tier III - Concurrently Maintainable (99.9%)
  - DC: 4-nines / Tier IV - Fault Tolerant (99.99%, assumes AC-equivalent DC equipment)

Results : Distributed Redundant

- **LIN System only achieves** 5-nines in the centralized system design
- **LIN + AC power infrastructure** achieves 5-nines at the facility
- **LIN + DC power infrastructure** achieves 4-nines at the facility

DR topology results in improved availability with increased CapEx and embodied carbon.

BR topology studied had lower CapEx with resulting lower availability results.

Key Design Consideration Findings

- LIN System is critical infrastructure
- Reliability results are driven by interdependencies between the liquid Nitrogen (LIN) system and the power systems.
- Distributed redundancy reduces blast radius impact as compared to block redundant (or catcher) systems.
- 800V DC data centre architectures depend on a fast-maturing and very different supply chain.
- MTTR reduction can improve availability.

Conclusions

- System availability is driven by system architecture i.e. block redundant vs distributed redundant etc. It is not individual components that define the overall system reliability.
- VEIR HTS solutions achieve five 9s of availability when integrated into appropriately designed system architecture.
- Achieving target availability requires a concurrently maintainable LIN system coupled with its associated power supplies. The rest is business as usual.

Call to Action

Work with the OCP community to develop optimal power distribution architectures for TIER III and TIER IV data centers using HTS conductors.

OCP HTS White Paper Workstream

OCP Power Distribution Project

VEIR HTS Solutions

Thank You!

